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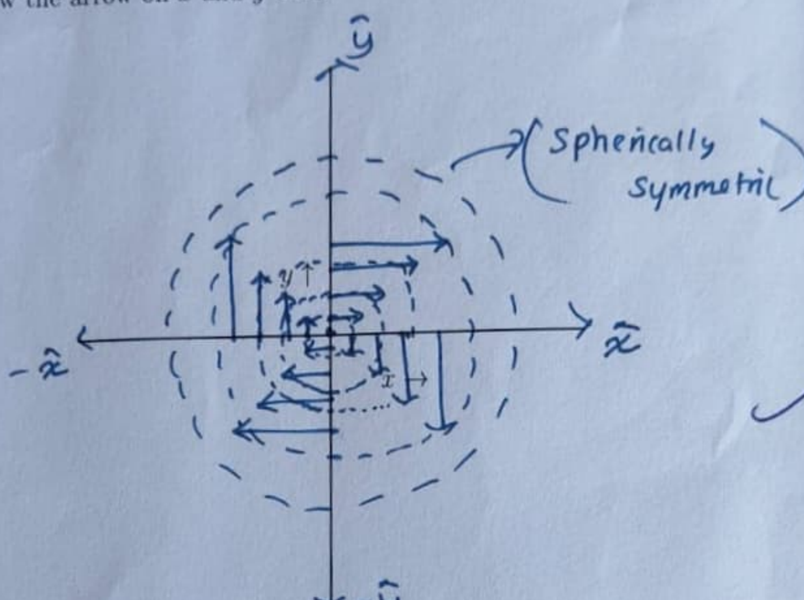
PHY102; Quiz 1; Inst.: Prof. Arvind; Dated : January 13, 2026

ROLL NO : MS NAME :

Max Marks=10 Time=15 minutes

Note : *Answers in the space provided or tick the box with correct option. For rough work use the blank side.*

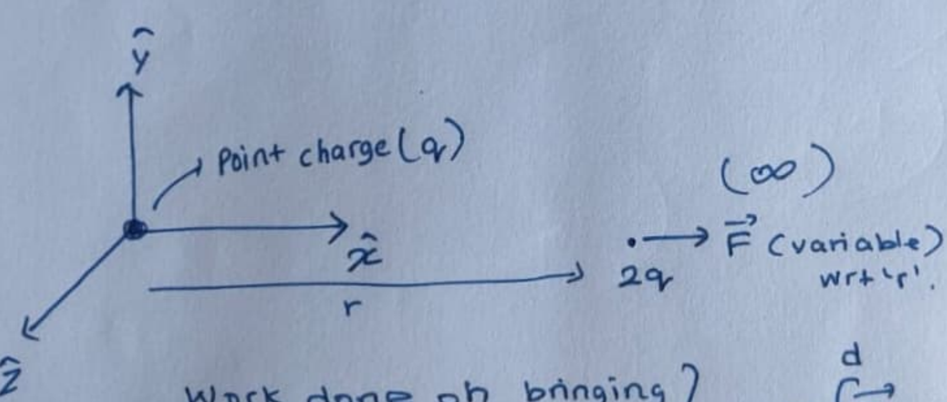
1. Draw the fields lines of the vector field $\vec{v}(x,y) = y\hat{x} - x\hat{y}$ in the x-y plane by drawing the vector arrow. Hint: First draw the arrow on x and y axes.



3

Marks=5

2. Consider a point charge q glued at the origin. Another point charge of magnitude $2q$ is brought from infinity to a distance d to this point charge along x axis. Calculate the work done in this process.



$$\underbrace{\text{Work done on bringing } 2q \text{ from } \infty \text{ to distance } d}_{W} = \int_{\infty}^{d} \vec{F}\cdot dr$$

$$W = -\int_{\infty}^{d} \frac{2q^2}{r^2}\,dr$$

$$W = -2q^2 \int_{\infty}^{d} r^{-2}\,dr$$

$$W = \left[\frac{-2q^2(-1)}{r}\right]_{\infty}^{d}$$

$$\boxed{W = \frac{2q^2}{d}}$$

in cgs units

or

$$\boxed{W = \frac{k\,2q^2}{d}}$$

in (SI)

Marks=5

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